Experiment - 2

ASM: Study of different types of Network Cables.

a) Understand different types of Network cable

Different types cable used in networking

1) Unshielded Twisted pair (UTP cable)
2) Shielded Twisted pair (STP) cable
3) Coaxial cable
4) Fibre optic cable.

b) Make you own Ethernet Cross-over cable / Straigh cable.

Tools and parts needed.

* Ethernet cabling is certified for gigabit support but cat 5 cabling works as well just over short distance.
* Two RJ45 plugs.
* Optional two plugs sheilds
* A Crimpling tool. This is all in one networking tool shaped to push down the pins in the plug and strip and cut the sheilding of the cable

Step 1 :- To start construction of the device begin by threading shields onto the cable.

Step 2 :- Next, strip approx 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

Step 3 :- After you will need to untangle the wires there should be four twisted pairs Referencing back to the sheet arrange them from top to bottom. One end should be in arrangement A and the other in B.

Step 4 :- Once the order is correct bush them together is line and if there are any that stick out farther than others snip them back create and even level.

Step 5 :- Next push the cable ~~is~~ right The ~~match~~ at the end of the plug needs to be just over the cable shielding and if it isn't that means that you stripped off too much shielding simple snip the cable back a.

Step 6:- After the wires are securely inside the plug insert into the crimping tool and push down.

Step 7:- Lastly repeat for the other end using diagram B (to make a crossover cabels) / using diagrams A (to make a straight through cable)

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	**Advantages** • Cheaper in cost • Easy to install as they have a smaller overall diameter. **Disadvantages** • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category6,6a	10Gbps	**Advantages** • Shielded. • Faster than UTP. • Less susceptible to noise and interference **Disadvantages** • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	
Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • **Disadvantages** • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	**Advantages** • High speed • High bandwidth • High security • Long distance **Disadvantages** • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	